

11.1 Constructing a Circumscribed Circle of a Triangle

Lesson Introduction

 Benchmark Focus	MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.		 Learning Targets	- I can construct the circumscribed circle of a triangle. - I can locate the circumcenter of a triangle.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How do you construct perpendicular bisectors of each side of a triangle? - How do you construct the circumscribed circle of a triangle? - How do you locate the circumcenter of a triangle?			
 Lesson Narrative	In this lesson, students discover that they can construct the circumscribed circle of a triangle by first constructing the perpendicular bisectors of the triangle and identifying the circumcenter. They discover that the circumcenter is equidistant from each of the vertices of the triangle. Then, they explore how they can place a compass on the circumcenter, adjust the length to reach each vertex of the triangle, and complete the circle.			
 Component Summary	11.1.1 Warm-Up (~5 min.)	Compare surface area and volume of lakes (<i>SE p. 177</i>)	11.1.2 Exploration (35-40 min.)	Discover how to construct the circumscribed circle of a triangle (<i>SE p. 178</i>)
	11.1.3 Wrap-Up (~5 min.)	Demonstrate mastery of lesson learning targets (<i>SE p. 181</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Inscribed Polygon in a Circle - Circumscribed Circle - Point of Concurrency - Circumcenter <u>Additional Glossary:</u> - Perpendicular Bisector		- Patty Paper - Compass - Straight Edge	
	<u>Additional Preparation</u> Prior to the lesson, work through 11.1.2 Exploration to ensure that you are familiar with the steps students will take and are prepared to support them as they work.			

 Teacher Tips	Common Misconceptions - Students may not understand that if a point is on the perpendicular bisector of a segment, then it is equidistant from the endpoints of the segment. - Students may not understand why the intersection of the perpendicular bisectors, the circumcenter, is equidistant from each vertex of the triangle. - Students may not understand how to use a compass and straightedge to construct perpendicular bisectors.	Things to Consider - In 11.1.2 Exploration, consider modeling the constructions and then having students practice independently a few times as they develop fluency with skills. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate when the circumcenter is inside/outside the triangle and on a side of the triangle.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect In 11.1.2 Exploration, foster students’ meta-awareness by prompting them to identify, compare, and contrast the locations of circumcenters in acute, obtuse, and right triangles. Support students’ meta-cognitive and meta-linguistic awareness as they share what the similarities and differences that they notice in the diagrams.	MA.K12.MTR.1.1: Participate in Effortful Learning 11.1.2 Exploration is a long and challenging activity. Foster perseverance in students by modeling the steps needed to perform the constructions, and recognizing their efforts during the activity. Develop students’ abilities to analyze and solve the problems by using the think aloud strategy to demonstrate expectations for student responses to the problems.
 Differentiate Instruction	In 11.1.2 Exploration, students are provided with a kinesthetic entry point as they use patty paper, compasses, and straightedges to construct perpendicular bisectors, a circumcenter, and a circle. Students are also provided with an auditory entry point as they discuss similarities and differences among the right, obtuse, and acute inscribed triangles. Provide a visual entry point by showing students images and definitions of obtuse, acute, and right triangles.	
Lesson Notes:		

11.1.1 Warm-Up: Lake Okeechobee and Deep Lake

Component Overview: Students compare the surface area and volume of Lake Okeechobee and Deep Lake to determine why it takes more Deep Lakes to cover the surface of Lake Okeechobee than to fill up Lake Okeechobee.



11.1.1 Warm-Up: Lake Okeechobee and Deep Lake

Lake Okeechobee means "big water" in the language of the Seminole tribe. Lake Okeechobee has a surface area of 730 square miles. Deep Lake in Big Cypress National Preserve is a naturally occurring sinkhole that was formed roughly 6,000 years ago. Deep Lake has a surface area of 0.002 square miles.

1. How many Deep Lakes are needed to cover the same surface area as Lake Okeechobee?

365,000 lakes

Lake Okeechobee averages 9 feet in depth, whereas Deep Lake averages 95 feet in depth. Based on the volume of the two lakes it would take approximately 30,850 Deep Lakes to fill up Lake Okeechobee.

2. Explain why it takes more Deep Lakes to cover the surface area of Lake Okeechobee than to fill up Lake Okeechobee.

Answers will vary. Surface area refers to the area of the water at the surface of the area (two-dimensional), while the volume includes the third dimension of height (in this case, depth). It takes more Deep Lakes to cover the surface of Lake Okeechobee than to fill it because Deep Lake is narrower but much deeper than Lake Okeechobee.



Consider sharing images, videos, or three-dimensional models to help students conceptualize the difference between surface area and volume. For example, you could show students a long thin graduated cylinder and a petri dish.

11.1.2 Exploration: Circumscribing A Triangle With A Circle

Component Overview: Students explore how to circumscribe a triangle with a circle through a series of scaffolded problems they complete with a partner. Students will need a compass, straightedge, and patty paper. Circulate while students work, to support struggling students and determine which students you will call on during the debrief discussion.



11.1.2 Exploration: Circumscribing A Triangle with A Circle



An **inscribed polygon in a circle** is a polygon which has all of its vertices on a circle.

$\triangle TWD$ is inscribed in Circle R , where $\overline{RW}$ is a radius. Lines RX , RY , and RZ are perpendicular bisectors of $\overline{TD}$, $\overline{TW}$, and $\overline{DW}$, respectively.

Each diagram shows a different type of triangle inscribed in Circle R .



Consider modeling the constructions and then having students practice independently a few times as they develop fluency with skills.

Acute	
Right	
Obtuse	

11.1.2 Exploration: Circumscribing A Triangle With A Circle (continued)

1. Determine similarities and differences among all three inscribed triangles.

Similarities	Differences
<i>Answers will vary. In all triangles, the three perpendicular bisectors intersect at one point.</i>	<i>Answers will vary. The perpendicular bisectors intersect in different places relative to the triangle (inside, on, outside the triangle)</i>

2. Point R is the center of the circle that is circumscribed about $\triangle TWD$. Make a conjecture about point R as it relates to the perpendicular bisectors of the sides of $\triangle TWD$.
Answers will vary. Point R is the intersection of the perpendicular bisectors.
3. Ask two other pairs of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of Their Conjecture	Initials
<i>Answers will vary.</i>	
<i>Answers will vary.</i>	

4. Review your conjecture with your partner and if necessary, make any changes.



A **circumscribed circle** is the smallest circle that includes a plane figure. If the figure is a polygon, then the circle must contain all of the vertices of the polygon. Not every polygon has a circumscribed circle, but all triangles and all regular polygons have circumscribed circles.

The **point of concurrency** is the point where three or more lines intersect.

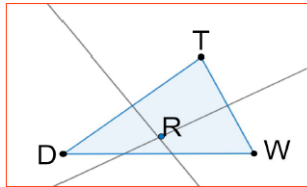


Compare and Connect

Foster students' meta-awareness by prompting them to identify, compare, and contrast the locations of circumcenters in acute, obtuse, and right triangles. Support students' meta-cognitive and meta-linguistic awareness as they share the similarities and differences they have noticed in the diagrams.

11.1.2 Exploration: Circumscribing A Triangle With A Circle (continued)

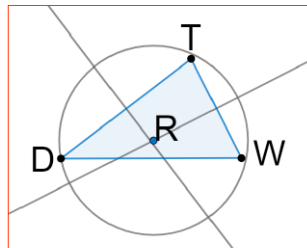
5. Copy $\triangle DTW$ onto patty paper. Construct the perpendicular bisectors of each side. Name the point of concurrency of $\triangle DTW$ point R .



6. Discuss with your partner the relationship point R has with each of the vertices.
Answers will vary. The point seems to be equidistant from each of the vertices.

The point of concurrency of the perpendicular bisectors of a triangle (the center of a triangle's circumscribed circle) is the **circumcenter**.

7. Use the definition of a circumcenter to explain how the relationship point R has with each of the vertices ensures that a circle can be drawn to circumscribe the triangle.
Answers will vary. Because each point is equidistant from the circumcenter, it will ensure that each vertex of the triangle will be on the constructed circle. The distance between the circumcenter and each vertex of the triangle is the radius of the circle.
8. Use a compass centered at point R to draw the circle that will pass through all three vertices.



Students are provided with a kinesthetic entry point as they use patty paper, compasses, and straightedges to construct perpendicular bisectors, a circumcenter, and a circle. Students are also provided with an auditory entry point as they discuss similarities and differences among the right, obtuse, and acute inscribed triangles. Provide a visual entry point by showing students images and definitions of obtuse, acute, and right triangles.



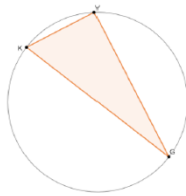
This Exploration is a long and challenging activity. Foster perseverance in students by modeling the steps needed to perform the constructions and recognizing their efforts during the activity. Develop students' abilities to analyze and solve the problems by using the think aloud strategy to demonstrate expectations for student responses to the problems.

11.1.2 Exploration: Circumscribing A Triangle With A Circle (continued)

9. Review your conjecture. Does your conjecture hold true for newly constructed circumscribed Circle R ?

Answers will vary.

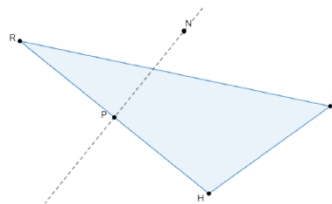
10. Copy $\triangle KYG$ onto another piece of paper. Then, use a compass and a straight edge to construct the circle that circumscribes $\triangle KYG$.



11. Describe how the circumscribed circle can be used to determine the type of triangle that $\triangle KYG$ is. Name a property and theorem from Unit 10 that can be used to verify this.

The circumcenter is the midpoint of the hypotenuse, the inscribed angle that corresponds to the arc intersected by the endpoints would be $\frac{1}{2}$ the arc measure or 90° . Therefore, the triangle is a right triangle by the Inscribed Angle Theorem.

12. The dotted line is the perpendicular bisector of side $\overline{RH}$. The distance between points R and N is 4.35 units. If point N is the circumcenter, what is the distance between points H and N ? Explain or show your reasoning.



If point N is the circumcenter, then $HN = 4.35$ units because the vertices of the triangles will be equidistant from the circumcenter.



Consider using a dynamic tool, such as GeoGebra, to enable students to investigate when the circumcenter is inside/outside the triangle and on a side of the triangle.



While circulating, watch for student responses to question 9. Their answers will provide insight into their understanding of questions 1-8. If necessary, debrief question 9 with a whole-class discussion.



After question 11, students will begin applying what they learned up until this point. Consider having a whole-class discussion after question 11 to ensure students are ready for the subsequent questions.

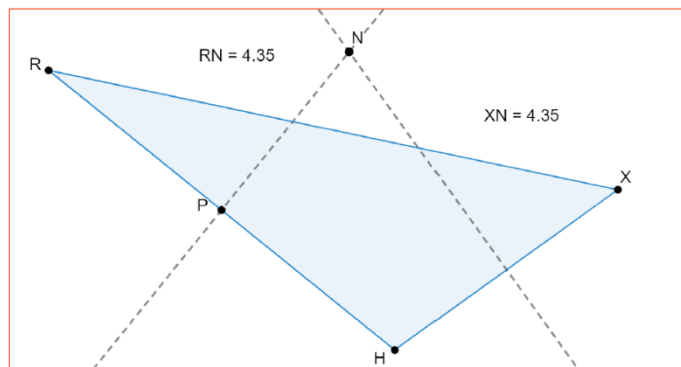


Students may have difficulty with question 12. Students may not label the 90° angles on the perpendicular bisectors. Consider prompting students who are stuck to articulate what information they know about triangle PNH .

11.1.2 Exploration: Circumscribing A Triangle With A Circle (continued)

13. Emilio tells his partner, Talia, that in order to show that point N is the circumcenter, he only needs to construct one more perpendicular bisector. Talia challenged him to prove it. Describe how Emilio could justify that point N is the circumcenter by drawing only one more perpendicular bisector.

If Emilio constructs another perpendicular bisector through $\overline{NX}$ (or $\overline{RX}$), he can determine if N is the circumcenter. If the second perpendicular bisector intersects with point N , then point N is the circumcenter.



11.1.3 Wrap-Up: Ticket Out the Door

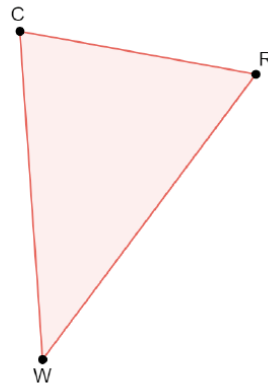
Component Overview: Students describe how to construct a circle that inscribes a triangle.



11.1.3 Wrap-Up: Ticket Out the Door

1. Describe how to construct Circle H , the circumscribed circle of $\triangle CRW$.

Construct perpendicular bisectors of two of the sides of $\triangle CRW$ and label the point of intersection as the circumcenter. Place compass on the circumcenter point and adjust the length to reach any vertex of the triangle. Then, use the compass to complete the circle that passes through all three vertices.



Use data from this formative assessment to determine which students need remediation before the next lesson.